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Palaeomagnetism of Palaeogene volcanic series of the Kamchatsky Mys Peninsula, East Kamchatka: the motion of an active island arc

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Abstract

The Paleocene Tarkhov and Eocene Baklan Formation are the lower and upper members, respectively, of a continuous island-arc sequence at the Kamchatsky Mys Peninsula on the Pacific side of the Kamchatka Peninsula. We studied Paleocene tuffs at five localities and Eocene tuffs and basalts at nine localities from beds of various attitude. Thermal demagnetization revealed consistent directions of the characteristic component (ChRM) of natural remanent magnetization in both formations. Normal and reversed ChRM directions in the Paleocene rocks are roughly antiparallel thus constituting a positive reversal test; at the same time, the fold test (McFadden and Jones, 1981) is not straightforward as locality-means display a banana-shaped distribution, most probably because of local rotations. Inclination-only data, however, show considerable decrease in dispersion after tilt correction thus constituting a positive fold test and implying a prefolding and most prprobably primary age of the ChRM in these rocks. The formation-mean inclination of $57.5 \pm 3.8°$ calculated with the aid of inclination-only statistics corresponds to a palaeolatitude of $38.1 \pm 4.1°$N. ChRM directions isolated from the Eocene Baklan Formation are well grouped and pass both the fold and reversal tests; the formation-mean inclination of $65.0 \pm 4.9°$ corresponds to a palaeolatitude of $47.0 \pm 6.4°$N. Geological data point to uninterrupted accumulation of the studied island-arc complex and, hence, to continuous activity of the subduction zone and related island arc. The difference between the Paleocene and Eocene palaeolatitudes of $8.9 \pm 6.3°$ is statistically significant, thus implying an absolute motion of the active subduction zone and related island arc. During the Paleocene to Eocene, no other active subduction zone existed between the studied area and Eurasia. We think that the studied island arc belonged to the Pacific plate and was moving northwestward, while the oceanic part of the Eurasian (or North American) plate was subducted southward underneath this arc; this island arc became inactive in the Late Eocene and was transported with the Pacific plate which started subducting under the Eurasia margin since that time.

Keywords: Kamchatka; palaeomagnetism; Palaeogene; tectonics; kinematics; North Pacific

1. Introduction

During the infancy of the plate tectonics conception, subduction zones were thought to be anchored in the mantle by descending slabs. About a decade ago, it was shown that the entire subduction zone may move with respect to the surrounding mantle and, therefore, to the Earth's rotation axis (Garfunkel et al., 1986). According to this model, the subduction zone should migrate towards the subducting

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plate and the migration velocity cannot exceed that of the overriding plate; for instance, the Aleutian island arc should move slowly oceanward.

As island arcs are attached to subduction zones and should migrate with them, palaeomagnetic studies of the island arc complexes can, in principle, be used to test this model. However, most modern island arcs belong to slowly moving, mainly continental plates, and displacements of the subduction zones lie within the error limits of the method. In addition, the subduction under many modern island arcs is nearly E–W in direction (e.g., the Equatorial Pacific) which virtually precludes the detection of these displacements with palaeomagnetic techniques.

There are remnants of numerous ancient island arcs now amalgamated into the active margin of the North Pacific. Palaeomagnetic data on these arc complexes are commonly used to establish their original positions and subsequent kinematics (Kovalenko, 1990, 1992; Bazhenov et al., 1992; Bol et al., 1992; Bol, 1993). If, however, an island arc complex exists for a long enough time interval, and the data are available from different levels of this complex, one can also evaluate the total displacement and velocity of the island arc due to migration of the related subduction zone. Recently, it was found that palaeomagnetic data on the Late Cretaceous–Early Paleocene island arc complex of the Kumroch Range (East Kamchatka) can be interpreted as movement of the active island arc (Pechersky and Shapiro, 1995; Pechersky et al., 1996). Here we explore the same subject with palaeomagnetic data on Palaeogene rocks of the Kamchatsky Mys Peninsula, still another ancient island arc complex of Kamchatka (Fig. 1).

2. Geological setting and sampling

The whole of Kamchatka is divided into a number of meridional-trending tectonic zones. The easternmost zone (East Peninsulas zone, EPZ) comprises the Kamchatsky Mys, Kronotsky (Fig. 1) and Shipunsky Peninsulas (not shown) on the eastern coast of Kamchatka where Paleocene–Eocene volcano-sedimentary rocks of island arc origin are wide-spread. This zone is regarded as an ancient island arc (Zinkevich and Tsukanov, 1992). At the same time, the northernmost of these peninsulas, Kamchatsky Mys, is regarded as the westernmost part of the Aleutians (Markov et al., 1972; Geits and Scholl, 1994). Still other researchers consider this peninsula as a separate exotic terrane (Watson and Fujita, 1985).

To the west of the EPZ, in the East Ranges zone (ERZ), another island-arc volcano-sedimentary complex is of Campanian–Maastrichtian age; the top of the section may be of Early Paleocene age (Tsukanov, 1991; Bakhteev et al., 1994). Overlying Paleocene and Eocene rocks are mainly sediments.

The above two tectonic zones are separated by the Tushev basin (Fig. 1) filled with folded late Cenozoic sediments. The eastern part of the Kamchatka Peninsula is thought to have formed from consecutive accretion of the Late Cretaceous–Early Paleocene Achayvayam–Valagina and Paleocene–Eocene Kronotsky island arcs (Tsukanov, 1991; Sokolov, 1992; Bazhenov et al., 1992; Zinkevich and Tsukanov, 1992; Shapiro, 1995). It is clear that the pre-accretional kinematics of these arcs are connected with the kinematics of the Northwest Pacific, mostly the Kula and Pacific plates (Engebretson et al., 1984; Kononov, 1989). At the same time, the co-existence of two active island arcs in the Early Paleocene gives reason to assume that a minor plate (or plates) had also existed in this region. The available palaeomagnetic data on the Kamchatka Cretaceous and Palaeogene island arc complexes show the latitudes much shallower than the Eurasian reference values, thus pointing to considerable shortening between these island arcs and the Eurasia margin (Kovalenko, 1990, 1992; Bazhenov et al., 1992; Gurevich and Surkis, 1994).

The timing of these movements is poorly constrained. On general geological grounds, it was hypothesized that the island arc displacement could have taken place simultaneously with the volcanic activity, and, therefore, the absolute displacement of the subduction zone (Shapiro, 1995). This hypothesis was confirmed by palaeomagnetic data on the Late Cretaceous–Early Paleocene island arc complex of the ERZ (Pechersky and Shapiro, 1995; Pechersky et al., 1996). Palaeomagnetic data on Eocene (Bazhenov et al., 1992) and Paleocene (Pechersky and Shapiro, 1995) rocks of the Kamchatsky Mys Peninsula appear to fit into this model too; however, the Eocene data are neither precise enough for a

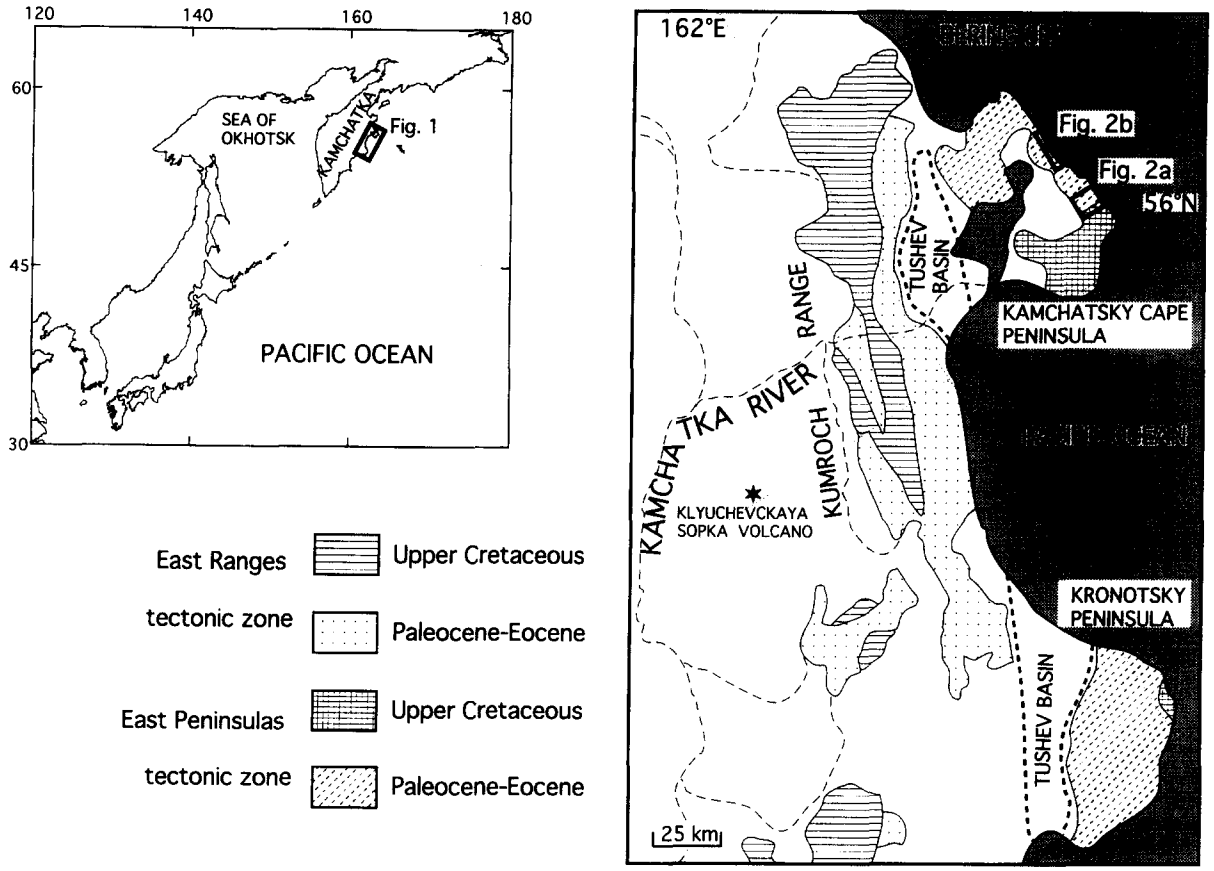


Fig. 1. Schematic geological map of East Kamchatka. Dot represents locality B9.

statistically significant conclusion nor confirmed by field tests. Below we present new data both on Paleocene and Eocene rocks of the Kamchatsky Mys Peninsula to test definitely the assumed absolute motion of the Palaeogene subduction zone and related island arc.

The Kamchatsky Mys Peninsula is divided into two domains separated by a major fault. The southern domain consists of several thrust sheets with complex internal structure and comprises strongly deformed Cretaceous sediments and volcanics and large bodies of gabbro and ultramafic rocks which are regarded as an ophiolite complex (Khotin, 1976).

The northern domain comprises more than 3-km-thick lower Palaeogene mostly volcano-sedimentary rocks of the Stolbov Series which is in turn divided into (from bottom to top) the Tarkhov, Vereschagin, Reef and Baklan Formations, with gradual transitions between them (Borzunova et al., 1969). This series accumulated on the slopes of a volcanic uplift, most probably an island arc. Basalt flows in the upper horizons of the Stolbov Series correspond to primitive island-arc tholeiites (Khubunaya, 1987). This succession of Maastrichtian to Late Eocene age (Beniamovsky et al., 1992) is deformed into large folds of northeast to meridional strike and cut by numerous faults. The precise age of folding is unknown as the series is overlain by Upper Pliocene rocks. However, the presumably Upper Oligocene–Lower Miocene conglomerates in the adjacent part of the Tushev basin contain numerous pebbles of rocks closely resembling those of the Tarkhov Formation, thus giving reason to assume that the early stages of deformation of the Stolbov Series are of Late Palaeogene age.

We studied the upper member of the Tarkhov Formation, consisting of intercalating olive-green re-washed fine-grained tuffs and siliceous rocks,

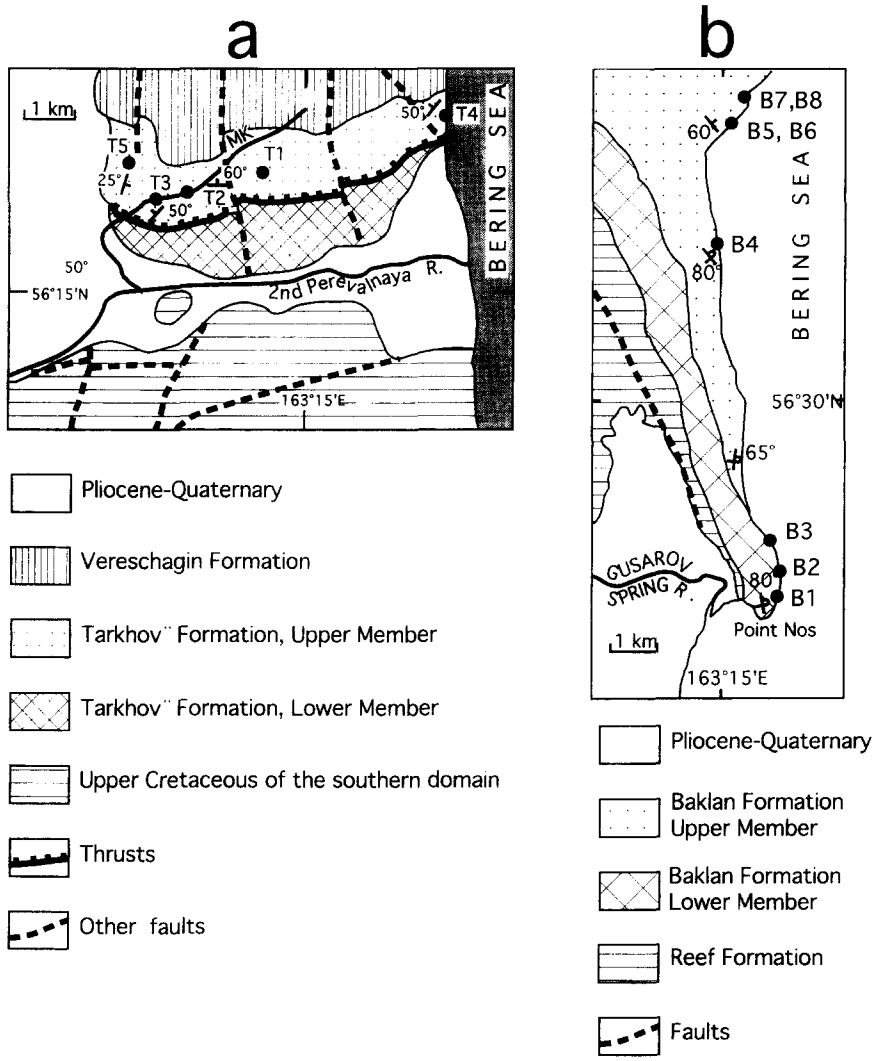


Fig. 2. Schematic geological maps of areas where the Tarkhov (a) and Baklan (b) Formations were studied with sampling localities shown as dots. *MK* (Fig. 2a) = Medvegy Kluch brook.

with sparse layers of basaltic gritstones and breccia. Judging by composition of debris and high content of quartz (up to 20%), the studied fine-grained tuffs correspond to andesite to dacite volcanism, whereas the overlying members of the Stolbov Series are mainly of basaltic composition. Benthic foraminifera (*Rhizamminain divisa, Bathysiphon nodosariaformis, B. akanosawensis, B. rzhehaki, Saccammina orbiculata, S. complanata, Ammodiscus glabratus, Cyclammina akkeshiensis, Bolivinopsis spectabilis, Rzhehakina lata, R. epigona, Silicosigmoilina futabaensis*) point to an Early Paleocene (61–65 Ma) age of the studied rocks (Beniamovsky et al., 1992). The most complete sections (up to 500 m thick) of the Tarkhov Formation upper member are in the southern part of a large isometric syncline in the valley of the Medvegy Kluch (Bear Spring) brook (Fig. 2a), in the northern part of the Kamchatsky Mys Range; this syncline is cut by numerous faults, sometimes with fault-conjugated short-wave folds. Oriented hand-samples of tuffs were taken at five localities from beds with various attitudes (localities T1–T5). At localities T1, T2, T4 and T5, the true thickness studied is several tens of metres

(Table 2); hence, we assume that the accumulation of rocks at each locality covered a significant time interval. At locality T1, a small subvolcanic intrusion of basalts of supposedly Early Palaeogene age was also sampled. The tuffaceous member of the Tarkhov Formation is separated by a gently dipping fault from a pile of basalts which are considered as the lower member of the same formation; unfortunately, the basalts lack any stratification and, hence, are not suitable for palaeomagnetic study.

The Baklan Formation is the youngest member of the Stolbov Series and, judging from foraminifera (*Textularia carinatoformis*, *Cyclammina pacifica*, *C. orbicularis*, *Recurvoides pseudoregularis*, *Cystominella grzybowskii*, *Robulus pseudovortex*, *R. antipodum*, *Dentalina catenula*, *D. laevigata*, *Nodosaria ewaldi*, *N. purula*, *Vaginulinopsis echinata*, *Pseudoglandulina inflata*, *Silicosigmoilina longa*, *Giroidina octocamerata*, *Cibicides hodgei*, *C. ventratumidus*, *Elphidium californicum*, *Nonion durhani*, *Bulimina debilis*, *B. curtissima*, *Trifarina advena californica*, *Globobulimina pacifica*, *Chilostomella balkanensis*, *Fissurina conscripta*) is of a middle to early Late Eocene age (36–49 Ma) (Beniamovsky et al., 1992); in particular, the basal horizons of this formation are of an early Middle Eocene age (43–46 Ma). Most samples were taken from a large northwest-trending syncline from the coastal exposures north of Cape Nos (Nose) (Fig. 2b). The lowest horizons of the formation represented by tuffaceous conglomerates with intercalations of finer-grained rocks were sampled from the overturned limb of the fold (localities B1 to B4). Stratigraphically above, a sequence of flysch-like intercalated tuffaceous sandstones and siltstones, with scarce gritstones, is exposed on the normal limb (loc. B5–B6). Earlier, the approximately same level was studied by Bazhenov et al. (1992) (loc. B7–B8).

The upper part of the Baklan Formation crops out at the extreme northeast of the peninsula where flat-lying (dips of about 1° on average) beds of sediments intercalate with a few flows of pillow-lava and are also intruded by sills and dikes which are thought to be nearly contemporaneous with sediments and lava. Here, samples were taken from two flows of pillow-lava (5 samples), one sill (4 samples) and six dikes (13 samples) of basalts and dolerites which will be further on treated as a single locality B9.

One hand-sample oriented with a magnetic compass was taken from each stratigraphic level and was then cut into several cubic specimens using a diamond saw. Therefore, the within-bed dispersion of palaeomagnetic directions could be estimated only if more than one specimen per sample was studied. When working on sedimentary rocks, the sampling points were more or less uniformly spaced across the studied interval, up to several tens of metres thick (Tables 2 and 3). Thus most localities cover fairly long stratigraphic intervals, and all sources of the between-bed dispersion was properly averaged.

3. Methods and procedures

The collection was studied in a similar manner in palaeomagnetic laboratories of the United Institute of Physics of the Earth and Geological Institute, both of the Russian Academy of Science, Moscow, Russia. One to three duplicate specimens from each hand-sample were subjected to stepwise thermal demagnetization in home-made ovens with the residual field of about 10 nT. All measurements of natural remanent magnetization, NRM, were made on Czechoslovakian spin-magnetometers JR-4 placed within large Helmholtz coils to reduce the ambient magnetic field. The bulk susceptibility was monitored during thermal demagnetization using a KLY-2 susceptibility bridge. Demagnetization data were analyzed with stereonets and orthogonal diagrams (Kirschvink, 1980). If more than one specimen was studied, the components isolated from each specimen were used for computation of sample-means which were in turn used for calculation of locality-means. Palaeomagnetic software written by Randy Enkin, Stanislav V. Shipunov and Trond H. Torsvik for IBM PC and by Jean-Pascal Cogné for Macintosh were used for analysis.

Three samples of basalts were studied on electronic microprobe CAMEBAX with accelerating tension of 15 kV and a current of 10 nA. Quantitative analysis was carried out for Ti, Fe, Mg, Mn, Cr and Al, with qualitative estimation for other elements. A probe of 1–2 μm in diameter was usually used; if the studied grains were heterogeneous, their average composition was measured with a probe of 10 to 20 μm in diameter. The Ti content and/or Ti/Fe ratio

were used for computation of the expected Curie temperatures.

In the earlier publication (Bazhenov et al., 1992), the result on the Baklan Formation was considered not very reliable being based mainly on end-points and not confirmed by any field tests. Duplicate specimens from this collection were re-studied in the Palaeomagnetic Laboratory, Institut de Physique du Globe, Paris, France. In total, 100 specimens from 54 hand-samples were thermally demagnetized in up to 14 steps and measured with a three-axes cryogenic magnetometer placed in magnetically shielded room. Demagnetization data were treated as the main collection.

4. Results of rock-magnetic studies

Unblocking temperatures T_b of the NRM in tuffs of the Tarkhov and Baklan Formations are grouped in the 480–540°C and 560–590°C intervals, the latter clearly prevailing. In contrast, lower T_b's (400–540°) are more common in basalts and dolerites, whereas temperatures of 550–580° are rare (Table 1). These different T_b's most probably correspond to two magnetic carriers which often co-exist in a sample. The same magnetic phases are visible on the curves of the SRM thermal demagnetization, but the low-temperature phase prevails in all rock types, particularly in magmatic rocks. Microprobe analysis shows that opaque minerals in basalts are represented by titanomagnetite with TiO_2 content ranging from 2.5 to 20 wt%. It is partly due to heterophase oxidation of titanomagnetite but also reflects the original variations of primary-magmatic titanomagnetites. Average Ti content of both homogenous and heterogeneous grains vary from 10 to 19 wt% and correspond to expected Curie temperatures between 400° and 225°, respectively. Such variations of titanomagnetite composition are characteristic in island-arc basalts (Pechersky et al., 1993). Other oxides were found in titanomagnetite grains too: mean values (ranges in parentheses) for twenty grains are: MgO 0.3% (0.1–0.6); Al_2O_3 1.6% (1–2.6); MnO 0.6% (0.0–2.2); Cr_2O_3 < 0.3%. The expected Curie temperatures are usually lower than the observed ones (Table 1), thus pointing to titanomagnetite decomposition. Optical observations revealed dissolution structures with lamellae more than 1 μm wide, which is typical high-temperature heterophase oxidation of titanomagnetite which leads to formation of secondary magnetite. The degree of oxidation is higher in tuffs than in magmatic rocks. This finding accounts for appearance of a magnetic phase with T_b's close to those of pure magnetite and the tuffs having higher coercivities than basalts and dolerites, above 30 mT and 15–17 mT, respectively. The viscous component accounts for a large part, sometimes up to 90%, of the NRM intensity in samples with high content of coarse-grained magnetite and titanomagnetite and, hence, high magnetic susceptibility.

5. Palaeomagnetic results

5.1. Lower Paleocene Tarkhov Formation

NRM intensity of the Tarkhov tuffaceous rocks vary widely (Table 2), both at the within-locality and between-locality levels. For instance, weakly magnetized varieties were found in interlayers up to 3 m thick at localities T1 and T2, whereas such varieties prevail at locality T3, with the only 30-cm-thick layer of high magnetic rocks at the site base. The Koenigsberger ratio (Q_n) in these rocks is usually above 2 and reaches 4.4 at site T3. Most samples are completely demagnetized by 600°C, thus pointing to magnetite and similar minerals as the NRM carriers. It is likely that coarse-grained magnetite predominates in samples with $Q_n < 0.5$ thus accounting for a three- to six-fold drop of NRM intensity of such samples after heating to 200° or 300°C. In contrast, the samples with a smaller intensity drop (less than 50%) often display lower magnetite concentration and relatively high Q_n values. This is likely to be related to diagenetic silicification and simultaneous acquisition of a chemical remanent magnetization. Irrespective of differences in demagnetization pattern, magnetic susceptibility of most samples does not change after heating to 400–450°C thus pointing to absence of maghemite, and sharp decrease of NRM intensity at low temperatures is most probably related to removal of the viscous remanence. We should add that demagnetization was often terminated as many samples literally exploded at intermediate temperatures, most probably due to zeolite disintegration. A small intrusion of basalts shows high magnetic susceptibility in the centre with a

Table 1
Rock-magnetic characteristics of representative samples of tuffs and basalts

Object	N	k,	NRM (A/m)	Q_n	ΔM	T_b (°C)
Tarkhov Formation						
Tuffs 1	17	0.56	0.017	0.84	0.59	583
		0.1–0.86	0.003–0.034	0.33–2.1	0.29–0.79	540–610
Tuffs 2	33	6.4	0.145	0.71	0.39	569
		1.19–28.1	0.02–0.515	0.22–2.6	0.17–0.94	510–600
Tuffs 3 (R)	11	2.63	0.112	0.86	0.69	567
		0.1–6.9	0.003–0.368	0.33–2.1	0.41–0.94	510–590
Tuffs 4	10	0.63	0.07	2.0	0.63	576
		0.1–2.6	0.003–0.335	0.4–4.2	0.46–0.75	540–590
Basalts baked margin	1	0.46	0.036	2.0	1.0	580
Basalts central part	3	65.8	0.7	0.28	0.41	580
		59.6–70.6	0.15–1.5	0.06–0.63	0.07–0.76	580
Baklan Formation						
Tuffs 5	33	8.54	0.18	0.39	0.71	520(14); 575(27)
		0.85–20.8	0.01–1.29	0.15–1.24	0.34–0.93	480–540; 560–590
Tuffs 6	12	20.0	0.18	0.19	0.50	528(5); 580(8)
		4.11–34.4	0.08–0.31	0.06–0.36	0.25–0.84	520–540; 560–580
Basalts 7 [a]	4	49.5	8.2	3.6	0.74	520(4) 580(1)
		26.4–61.3	7.1–126	3.4–119	0.5–0.93	500–530
Basalts 8	18	50.1	0.9	0.47	0.6	516(15); 565(11)
		34.2–77.3	0.23–3.5	0.1–1.3	0.1–1.46	400–530; 550–580

[a] Aa sample with anomalously high value of $Q_n = 119$ is excluded from computation of mean values of Q_n and NRM.
Comments. Object: 1 = low magnetic tuffs from localities T1 and T2; 2 = more magnetic tuffs from the same localities; 3 = reversely magnetized tuffs from the lower parts of the section at localities T1 and T2; 4 = tuffs from locality T3; 5 = tuffs from localities B1–B6 with prefolding component; 6 = tuffs from the same localities with unstable and/or anomalous component directions; 7 = lava flows; 8 = intrusives, mostly dikes; N = the number of hand-samples studied; k = magnetic susceptibility (in 10^{-3} SI units); NRM = intensity of natural remanent magnetization; Q_n = Koenigsberger ratio; ΔM = the ratio of magnetization intensity after heating to 300°–400°C to that of the NRM; T_b = unblocking temperature of the NRM. Mean parameter values and corresponding ranges are given for each object in the first and second lines, respectively.

sharp drop towards the margins (Table 1) which is typical for these rocks. Low temperature components are well defined in this basalts, showing low directional scatter, whereas high temperature components are quite dispersed; the high-temperature data were discarded.

After removal of an unstable remanence at 200–250°C, the NRM in most samples of grey tuffaceous rocks may be accounted for by a single characteristic component (ChRM) (Fig. 3a–d). Orthogonal plots may show the rectilinear decay to the origin or only slightly miss it, but the ChRM direction, on average, remains the same irrespective of whether the adjusted linear segments pass through the origin or not.

In some samples, however, another component appears to be present at intermediate temperatures (Fig. 3e–g). This intermediate temperature remanence is much more dispersed than the ChRM's, and its directions fall between the corresponding ChRM directions and that of the present-day field. Although sometimes represented by rectilinear segments on demagnetization plots, this component is hardly a true component but results from partial overlapping of unblocking spectra of a recent overprint and the ChRM. It should be added that many samples with strong initial drop of the NRM intensity showed erratic behaviour after heating of 350–450°C, this being the main cause for the data rejection.

The ChRM persist up to 560–580°C, sometimes to as high as 600°C. Judging by unblocking temperatures, the main ChRM carrier is magnetite; as for higher unblocking temperatures, we attribute them to a partial oxidation of magnetite which may lead to an increase of Curie temperature (Nguyen et al.,

Table 2
Paleomagnetic results from the Tarkhov Formation

S	N	T (m)	A/d	In situ				Tilt-corrected			
				D	I	k	α_{95}	D	I	k	α_{95}
1	39/24	31	1/63	155.1	54.1	26	5.6	28.9	57.2	27	5.5
2	20/13	9	351/66	141.3	40.8	24	7.9	39.8	59.6	25	7.8
3	18/14	3	312/49	84.3	51.3	25	7.5	8.1	56.1	23	7.8
4	22/15	155	309/52	92.6	55.9	4	17.3	350.6	61.2	12	10.7
5	15/12	90	107/25	30.6	66.0	18	9.5	66.2	53.2	48	5.8
Overall mean											
78 samples				117.6	60.4	7	6.2	26.8	59.8	15	4.1
5 locality means				109.8	60.8	9	20.8	28.0	60.2	27	12.0
Inclination-only statistics (McFadden and Reid, 1982):											
78 samples				–	51.9	16	4.4	–	*57.5*	*23.8*	*3.8*
5 locality means				–	54.6	10	26.7	–	57.6	244.1	3.2

S = locality number as in the text and Fig. 2a; N = the number of samples studied/accepted; T = true thickness studied; A/d = mean azimuth of dip/angle of dip; D = declination; I = inclination; k = precision parameter; α_{95} = radius of 95% confidence circle. All directions are inverted to normal polarity. The results printed in italics are used for interpretation. Fine-grained tuffs were sampled from all localities. At locality T1, a small subvolcanic intrusion of basalts was also studied.

1978). Neither polarity nor direction of the ChRM depend upon sample lithology and rock-magnetic characteristics.

Both polarities of the ChRM are found, and the number of samples with the ChRM of normal and reversed polarities is nearly the same (Fig. 4a). Both polarities were found at localities T-1 and T-2, with some transitional directions between the zones of different polarity. These 'transition zones' are found at both the upper and lower boundaries of the reversed interval at locality of T1 zone (Fig. 5) and at its upper boundary at site T-2. After omission of the transitional directions, tilt-corrected ChRM unit vectors of both polarities are nearly antipodal (Fig. 4a).

The ChRM directions are well grouped at each locality, and the corresponding locality-means are well defined (Table 2). The dispersion of locality-means is reduced after tilt correction (Fig. 4b–c), but the fold test (McFadden and Jones, 1981) is not conclusive. Closer inspection showed that mean inclinations for all the localities are statistically indistinguishable after tilt correction, but the declinations differ and are distributed along a small circle centred at the projection pole (Fig. 4c). We assume that the dispersion of declinations results not from a remagnetization, but from net rotations of smaller blocks (MacDonald, 1980) which is in accord with geological data which point to a rather complicated deformation of the study area (Fig. 2a). Moreover, the declinations significantly correlate with the strikes (Table 2) and this also points to the local rotations as the cause of declination scatter. These rotations also account for the fact that the reversed unit vectors are more scattered than the normal ones (Fig. 4a), as only the reversed polarity of the ChRM was found at most rotated localities, T4 and T5. Thus, the magnetization is considered to be of prefolding age.

The inclination-only statistics (McFadden and Reid, 1982) was applied to the data, both at the sample and locality-mean levels: thus obtained mean inclination is $57.5 \pm 3.8°$. A significant improvement in data grouping, especially at the locality-mean level (Table 2), points to the prefolding age of the ChRM in these rocks. Moreover, the positive fold and reversal tests as well as the presence of transitional directions at the boundaries of the normally and reversely magnetized parts of the section (Fig. 5) imply the primary age of the ChRM of the Tarkhov Formation.

5.2. Eocene Baklan Formation

Magnetic susceptibility of the Baklan tuffs is rather uniform and much higher than that of the

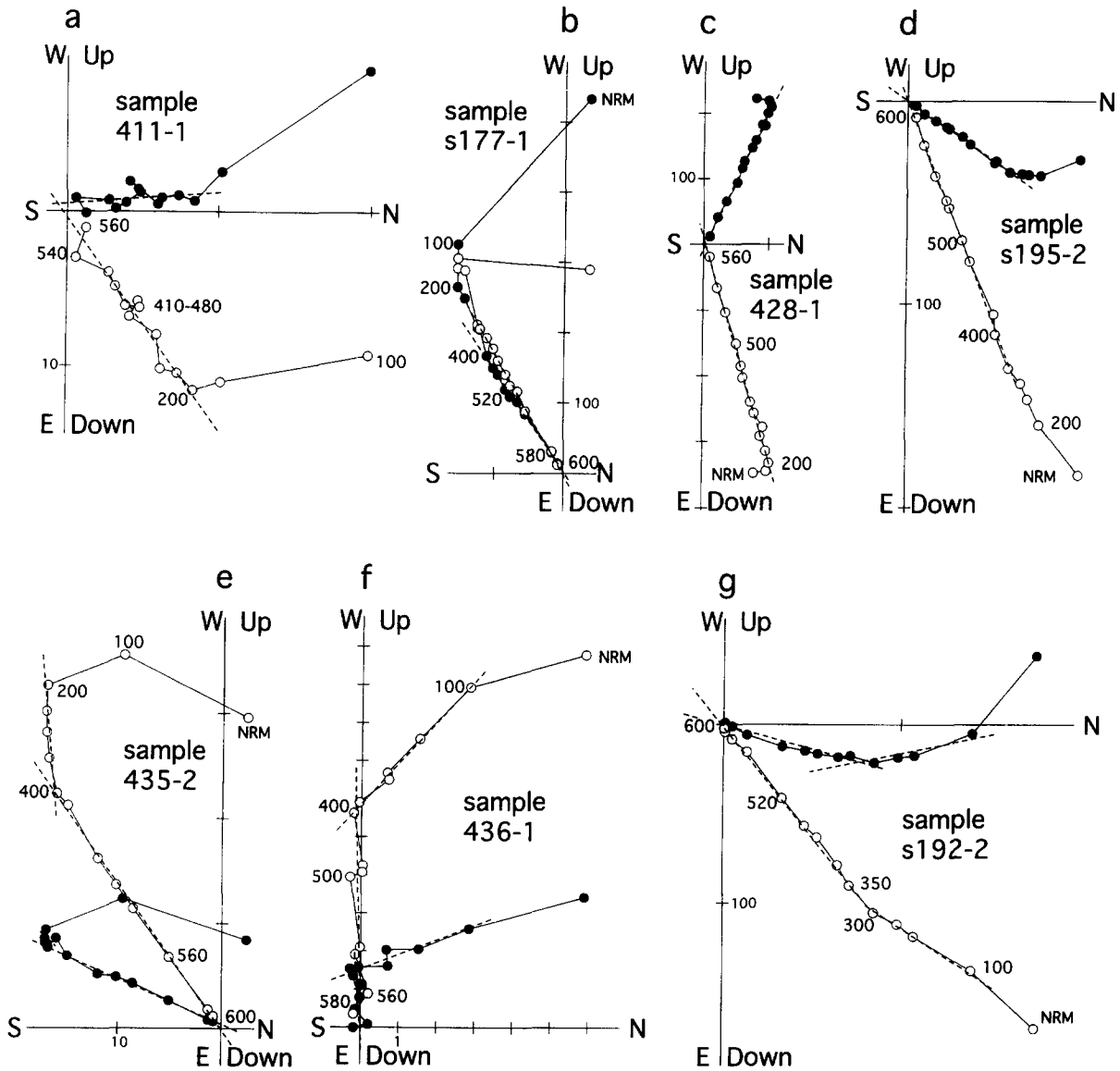


Fig. 3. Representative orthogonal plots for tuffaceous rocks of the Paleocene Tarkhov Formation. Solid (open) symbols are projected onto horizontal (vertical) plane. The plots are in stratigraphic coordinates. Isolated components are shown as dashed lines. Temperatures are in degrees Celsius.

Tarkhov rocks, whereas their NRM intensities are relatively lower. Q_n values are, on average, about two times lower too (Table 1). The difference in susceptibilities is most reasonably explained by the predominately basaltic composition of the Baklan Formation and the more silicic composition of the Tarkhov rocks. Lower NRM intensities and Q_n values may be attributed to the detrital origin of the NRM and relatively larger share of coarse-grained magnetite and titanomagnetite in the former rocks. The latter is confirmed by the fact that the samples with anomalous component directions yield the noisiest demagnetization data and have the highest susceptibilities, up to 34×10^{-3} SI units, and lowest Q_n values. The samples of this type were excluded from computation of the formation-mean. Note, however,

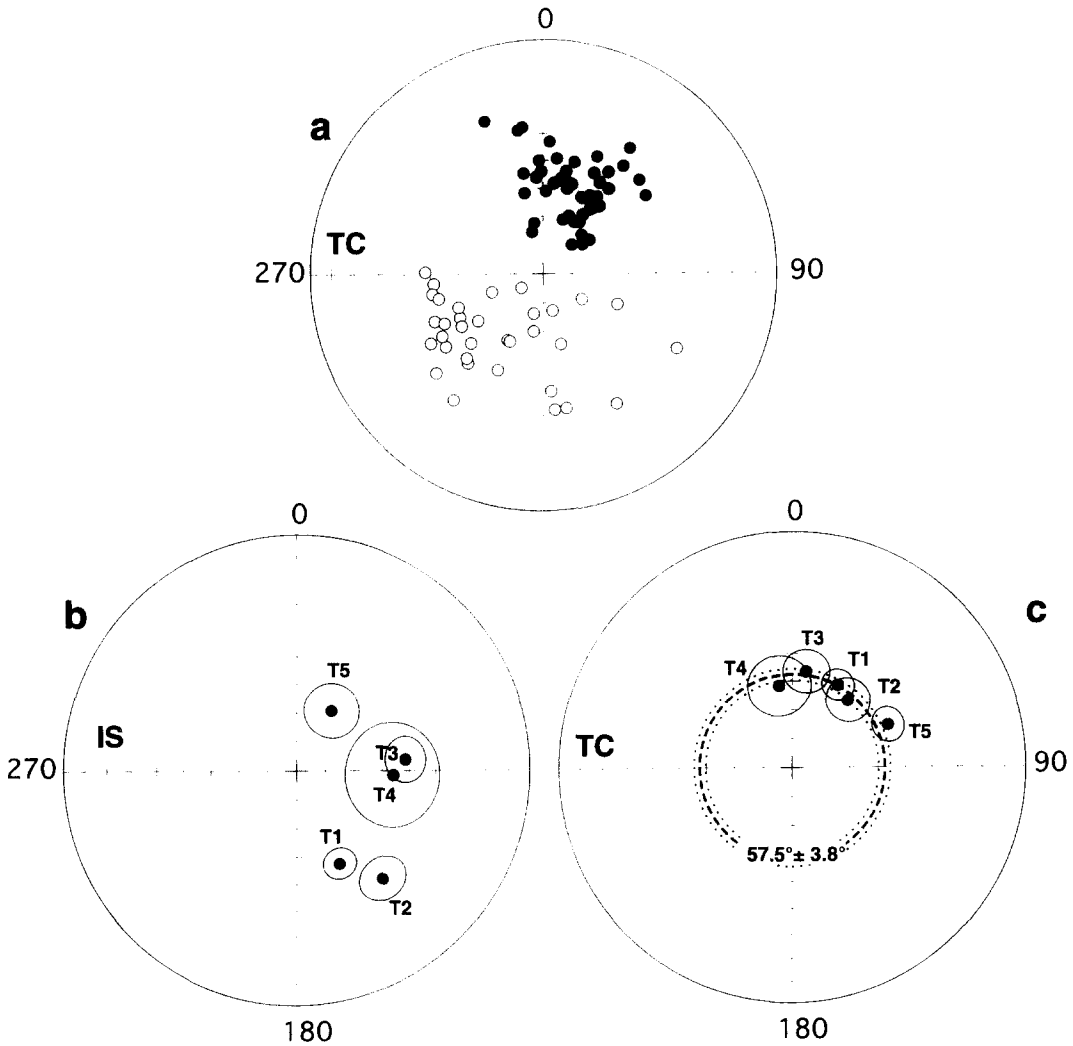


Fig. 4. Stereoplots of ChRM directions from the Paleocene Tarkhov Formation. (a) Distribution of sample-mean directions after tilt-correction. Solid (open) dots are projected onto lower (upper) hemisphere. (b, c) Distribution of locality-mean directions with associated confidence circles in situ (b) and after tilt correction (c): all directions are inverted to normal polarity. Thick dashed line is mean inclination calculated with the aid of inclination-only statistics (McFadden and Reid, 1982). Thin dotted lines denote confidence limits of mean inclination. All symbols are projected onto the lower hemisphere.

that the entire collection, irrespective of the sample behaviour during demagnetization, has very similar unblocking temperatures (Table 1). Thus the magnetic carriers are similar everywhere, and only their concentration and grain size vary.

After removal of a viscous overprint, some samples of tuffaceous rocks revealed a single characteristic component showing the rectilinear decay to the origin or just slightly missing it (Fig. 6a–d). An intermediate-temperature component, however, was recognized from 300° to 500°C in many samples (Fig. 6e, f); after its removal, a high-temperature component was isolated above 500–540°C and up to 560–580°, sometimes as high as 610°C. The ChRM's from the single-component samples and high-temperature component in samples of the latter type are similar, and we presume that it is the same remanence. Directions of the intermediate-temperature component are much more dispersed and usually lie near to great circles between the corresponding

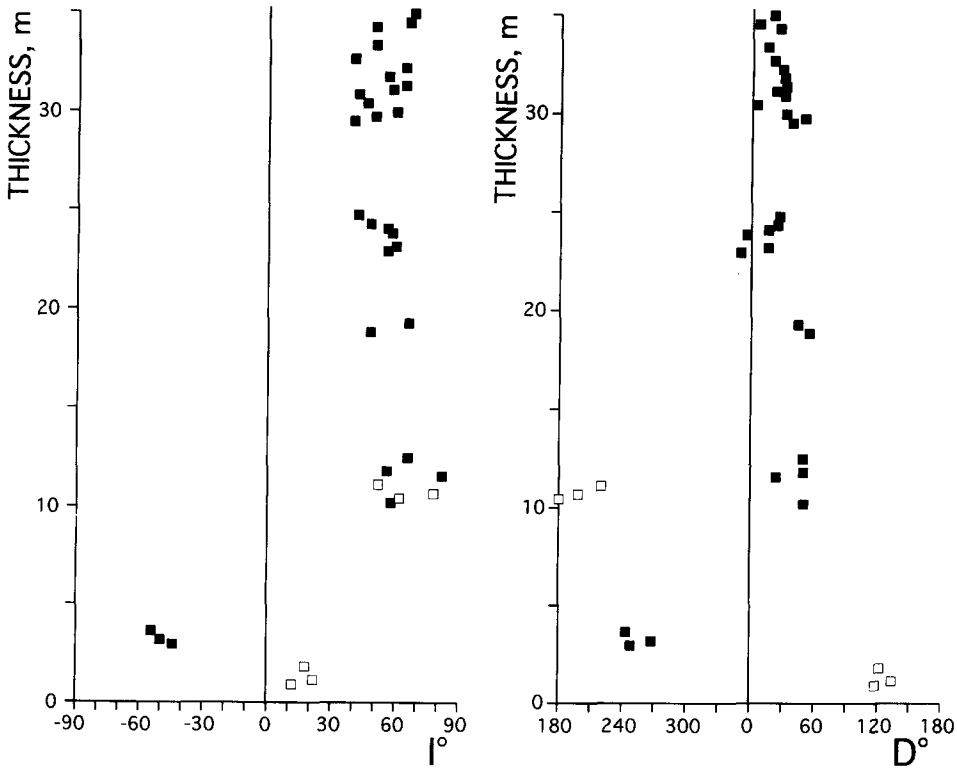


Fig. 5. Distribution of tilt-corrected ChRM declination (*D*) and inclination (*I*) through the studied stratigraphic section of the Tarkhov Formation (locality T1). Transitional directions are shown as open squares and were not used for calculation of locality-mean directions in Table 2. Large gaps between the sampling points correspond to coarse-grained and/or fractured rocks.

ChRM directions and that of the present-day field. In the same manner as for the Tarkhov Formation, we attribute this pattern to partial overlapping of unblocking spectra of the recent overprint and the ChRM and excluded the intermediate quasi-component from consideration. Finally, some samples displayed erratic behaviour above 450–470°C, and only the intermediate-temperature component could be isolated (Fig. 6h). Such samples often have high susceptibilities, low Q_n values and display sharp drop of the NRM intensity after heating to 200° or 300° which may be accounted for by a larger share of coarse-grained magnetite and titanomagnetite.

The observed unblocking temperatures confirm the rock-magnetic data that the main ChRM carriers are titanomagnetite and magnetite in spite of considerable variations in behaviour during the treatment. As for the higher unblocking temperature, we again attributed them to a partial oxidation of magnetite. Rock-magnetic characteristics are also very similar throughout the entire collection (Table 1), thus implying that the same ferromagnetic minerals are present everywhere, and that the differences in the NRM demagnetization are due to variations of their concentration and grain size. In a previous study, a collection of the Baklan tuffs from the overturned limb was rejected because of chaotic dispersion of palaeomagnetic directions (Bazhenov et al., 1992); the re-study of the duplicate specimens just confirmed the earlier finding. Exactly the same pattern was observed at site B3. We cannot propose an unambiguous explanation of such a pattern.

Volcanics of this formation (site B9, Fig. 1) have high susceptibilities but relatively low Q_n values, less than 5 in lava flows and less than 1.5 in sills and dikes (Table 1). Rock-magnetic characteristics and NRM unblocking temperatures ranging from 400° to 580°C imply titanomagnetites as the main NRM carriers. Titanomagnetites are nearly primary in some samples but more often they are oxidized, some-

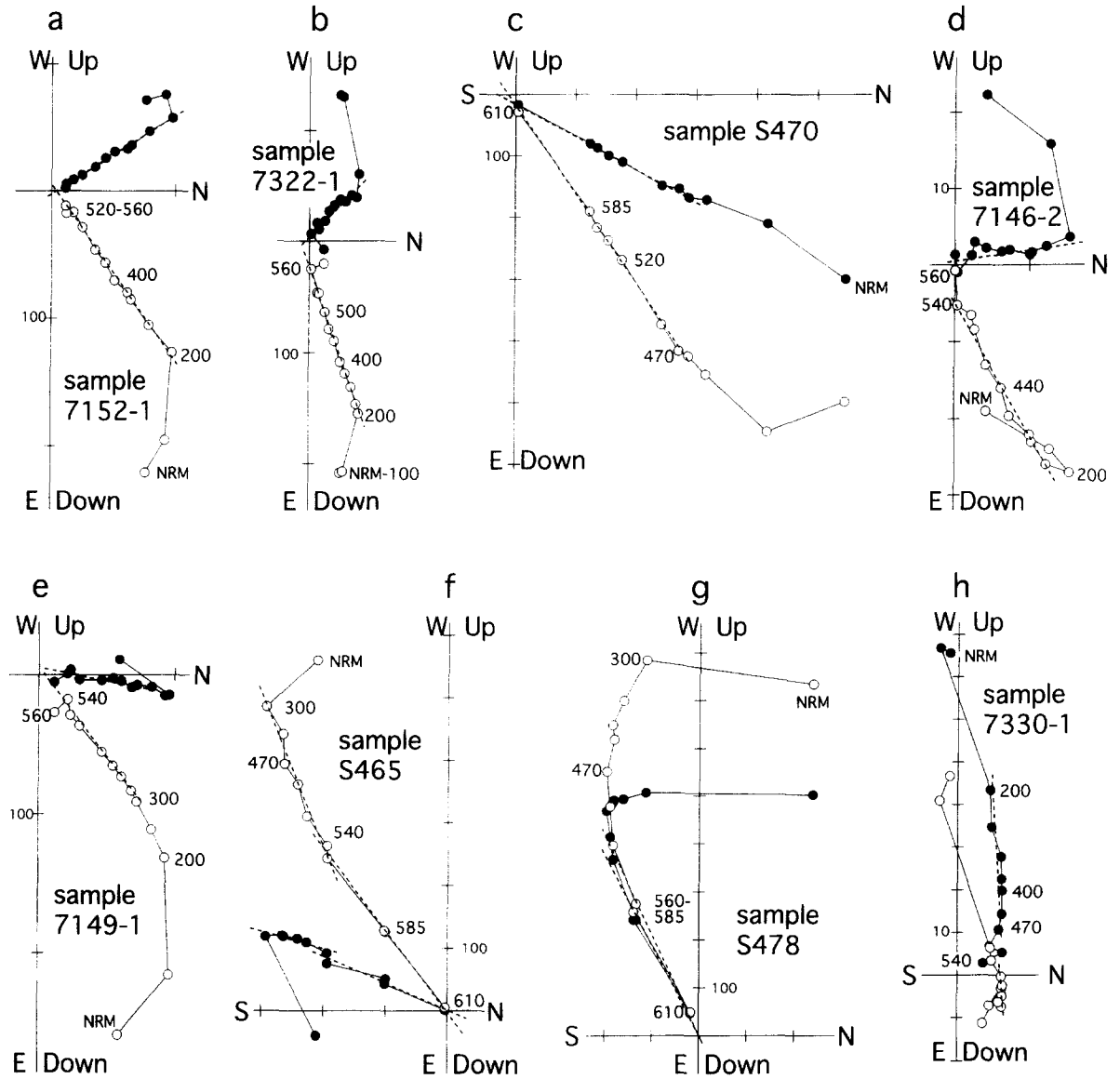


Fig. 6. Representative orthogonal plots for tuffaceous rocks of the Eocene Baklan Formation. Notation as in Fig. 3.

times to pure magnetite. Several magnetic phases with different Curie temperatures may co-exist in a sample.

Some samples of volcanics from site B9 were unstable during demagnetization, whereas others revealed well-defined single characteristic components (Fig. 7a–c). In two more samples, two components with antiparallel directions were isolated (Fig. 7d–f). The ChRM's in five dikes out of six studied are well grouped with nearly antiparallel directions of both polarities (Fig. 8a). Mean direction of this bipolar component is well defined and close to the tilt-corrected data on tuffs (Fig. 8c, Table 3). In contrast, the ChRM in basalt flows and sills are either nearly chaotic, usually for less thermally stable samples, or yield data, well grouped within a unit, but directionally anomalous with respect to the data from dikes and sedimentary rocks.

It may be that the nearly antiparallel components in some samples may result from a self-reversal,

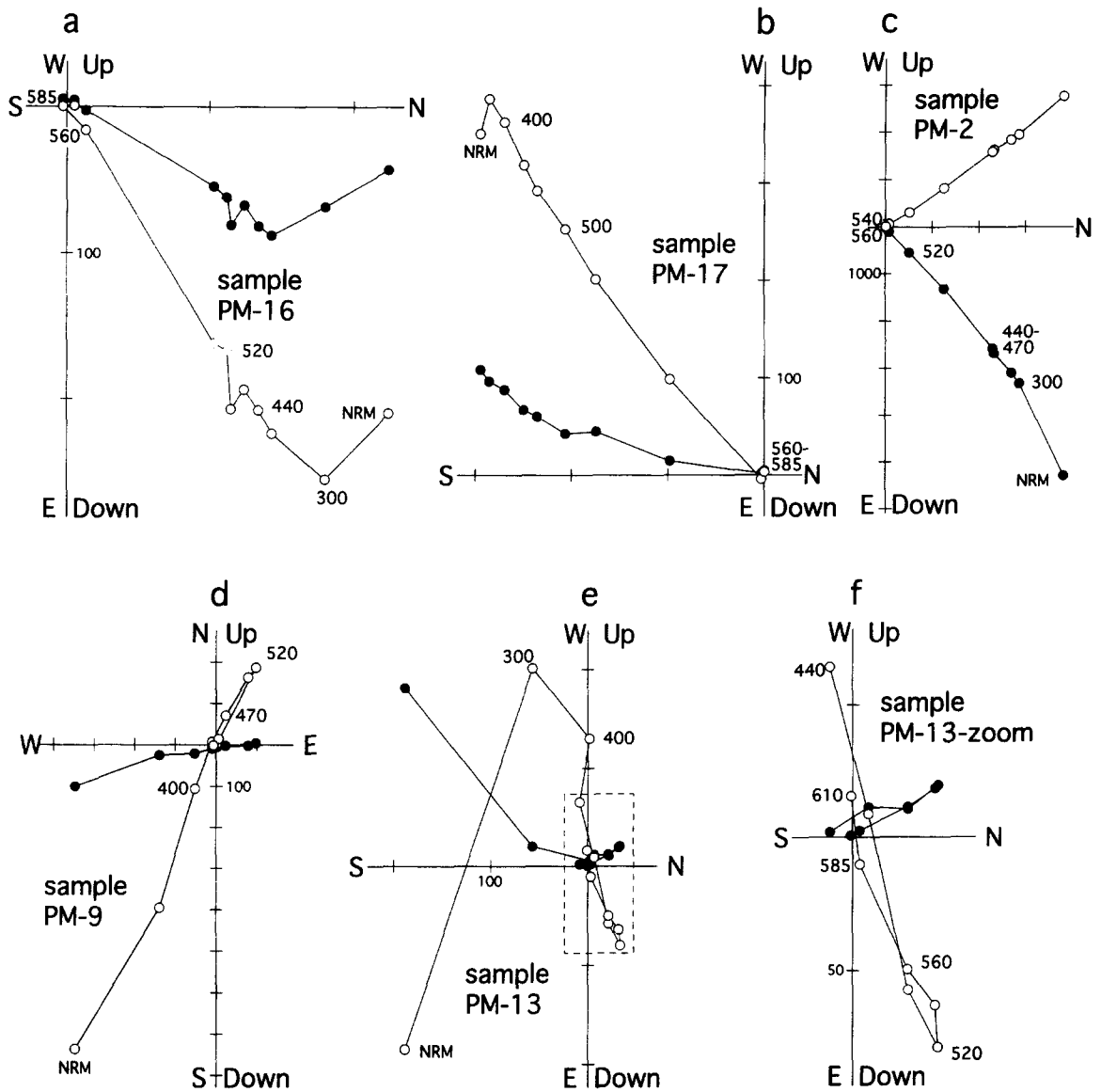


Fig. 7. Representative orthogonal plots for basalts of the Eocene Baklan Formation. Notation as in Fig. 3. Note: plot (f) is an expanded view of plot (e).

in accord with the co-existence of several magnetic phases. At the same time, the presence of two polarities in dikes and well-clustered but directionally anomalous directions in other volcanic bodies implies that magmatic activity could overlap a polarity transition. It thus may be that the antiparallel components may record a quick switch of the field. In any case, the perfect antiparallelity of normal and reversed directions points to an ancient age of the ChRM at this locality in spite of the close agreement of the site-mean vector and the present-day field direction.

With the only exception of the above-mentioned locality B3, the ChRM directions from the other localities are well grouped, and locality-means are well defined. Note that the re-studied collection (localities B7, B8) yields results very close to those from other localities and the previously reported direction

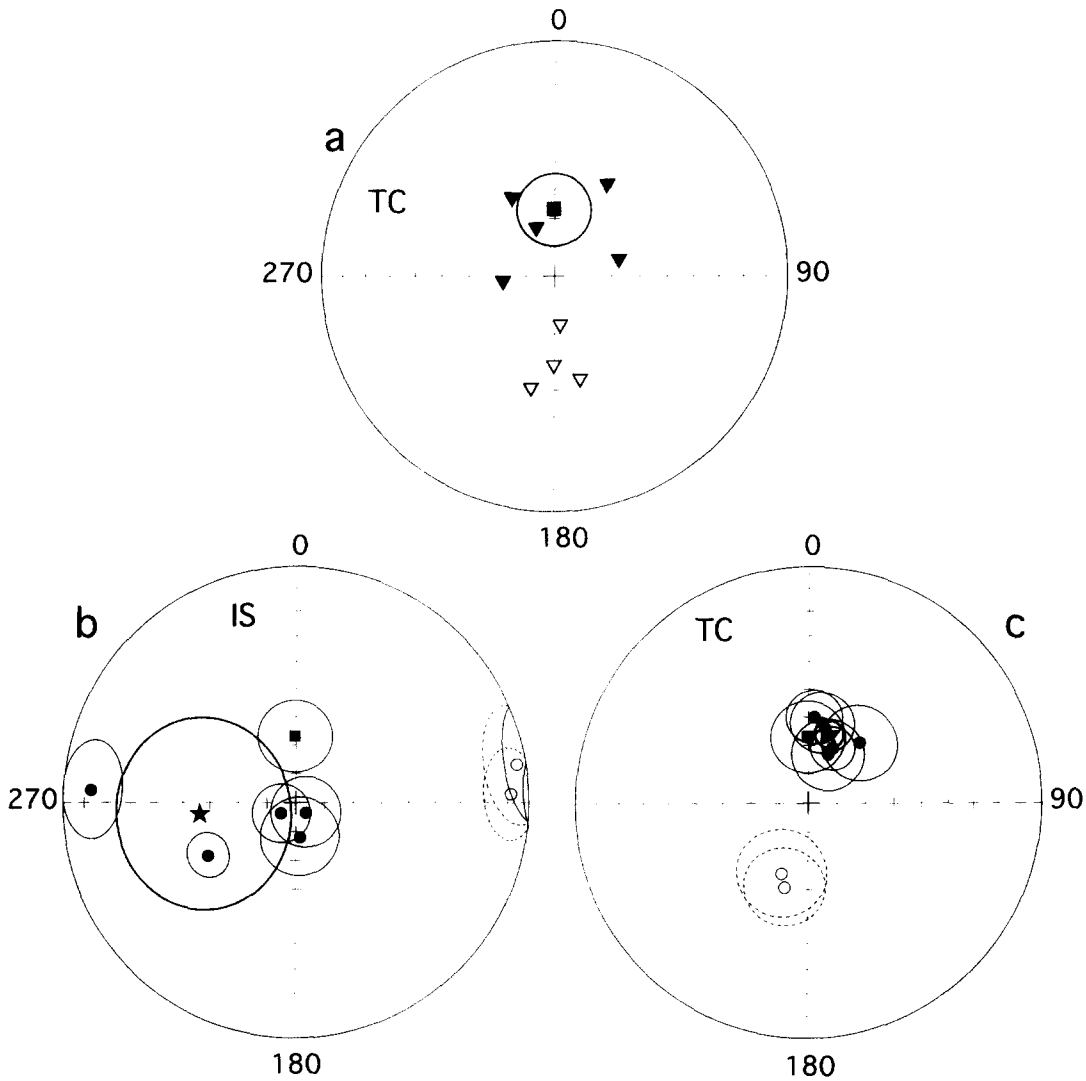


Fig. 8. Stereoplots of ChRM directions from the Eocene Baklan Formation. (a) Distribution of unit directions after tilt-correction in basalts (locality B9). (b, c) Distribution of locality-mean directions in tuffaceous rocks (dots) and basalts (square) with associated confidence circles in situ (b) and after tilt correction (c). Star and thick line are overall mean and associated confidence circle, respectively. Solid (open) symbols and solid (dashed) lines are projected onto the lower (upper) hemisphere.

(Bazhenov et al., 1992). Two polarities are present, and the reversal test for the tilt-corrected directions is positive (Fig. 8c, Table 3). Locality-means cluster much better after tilt correction (Fig. 8b, c, Table 3), and the fold test (McFadden and Jones, 1981) is positive. In addition, the plot of palaeomagnetic directions versus the studied section of sedimentary rocks shows some anomalous directions between the normally and reversely magnetized intervals (Fig. 9); these directions may be a record of the polarity transition. Thus the positive fold and reversal tests and the presence of transitional directions between the normally and reversely magnetized parts of the section imply a primary age of the ChRM of the Baklan Formation.

6. Interpretation and discussion

We isolated dual-polarity prefolding and presumably primary components from the lower and upper

Table 3
Paleomagnetic results from the Baklan Formation

S	N	T (m)	A/d	In situ				Tilt-corrected			
				D	I	k	α_{95}°	D	I	k	α_{95}
B1	8/7	55	65/101	87.7	−8.9	19	12.1	195.7	−59.6	15	13.8
B2	9/8	16	63/105	80.3	−4.5	11	15.3	200.7	−63.9	10	15.4
B4	4/4	4	66/91	273.5	13.1	27	13.4	3.9	59.9	55	9.5
Mean for overturned limb											
19 samples				265.9	8.0	5	8.3	14.9	61.6	15	8.2
3 locality means				267.1	8.9	8	12.0	13.1	61.3	287	7.3
B5	9/6	11	46/55	238.9	54.1	59	7.5	23.7	69.2	67	7.0
B6	9/9	8	26/37	173.2	77.9	12	13.6	40.4	62.4	13	13.1
B7	17/9	70	18/33	234.8	83.7	21	10.1	9.8	61.6	19	10.8
B8	17/11	30	9/21	135.2	85.0	12	12.2	21.7	71.8	12	12.5
Mean for normal limb											
35 samples				211.8	80.1	12	6.9	23.8	66.8	15	6.0
4 locality means				218.5	77.9	22	20.0	24.0	66.7	120	8.4
Overall mean for sedimentary rocks											
54 samples				253.8	58.0	4	9.5	*20.3*	*65.0*	*15*	*4.9*
7 locality means				257.2	50.2	4	34.3	18.8	64.5	132	5.3
B9	22/9		0/1	359.4	67.0	17	11.2	359.4	66.9	17.4	11.2

Tests
Reversal test
$F_{(2,122)} = 3.07$; tilt-corrected: $f = 0.99$
Fold test
$F_{(14,110)} = 1.78$; in situ: $f = 25.44$; tilt-corrected: $f = 0.90$

S = localities as in text and in Fig. 2; F = 95% critical value of F statistics with the number of degrees of freedom in parentheses; f = calculated values of F statistic. The results printed in italics were used for interpretation. Other notation as in Table 2. Fine-grained tuffs, tuffaceous sandstones and siltstones were sampled at localities B1 to B8. Samples were also taken from two flows of pillow-lava, one sill and six dikes of basalts which are treated as locality B9 (see text for more detail).

members of the island-arc Stolbov Series. Either the Eurasian or North American Early Tertiary reference values of about 67°N (Besse and Courtillot, 1991) are much higher than the observed Paleocene and Eocene palaeolatitudes of $38.1 \pm 4.1°N$ and $47.0 \pm 6.4°N$, respectively (Fig. 10). This finding implies that the studied island arc was originally more than 20° of latitude to the south from the Eurasian margin.

Also very important is the fact that the formation-mean inclinations differ significantly by $7.5 \pm 5.0°$ (Tables 2 and 3); this corresponds to a latitudinal difference of $8.9 \pm 6.3°$. Mean inclinations for different levels of the Baklan Formation also appear to become steeper with decreasing age but these differences are statistically insignificant (Table 3, Fig. 10). Thus there is a systematic steepening of inclination (= change of palaeolatitudes) from bottom to top within the studied section. We attribute this finding to the northward drift of the Kronotsky island arc from the Early Paleocene to early Late Eocene.

Palaeontological data (Beniamovsky et al., 1992) point to an Early Paleocene age (61–66 Ma) of the studied upper member of the Tarkhov Formation. The basal horizons of the Balkan Formation at localities B1 to B8 are of an early Middle Eocene age (43–46 Ma), whereas the age of host rocks at locality B9 where both polarities were found is likely to be 36–40 Ma (Beniamovsky et al., 1992). Taking into account the large difference in ages of the studied parts of the Baklan Formation, we prefer not to combine them. Thus the accumulation of the Stolbov Series island arc took place from the Paleocene up to the early Late Eocene, i.e. for about 30 Ma, whereas

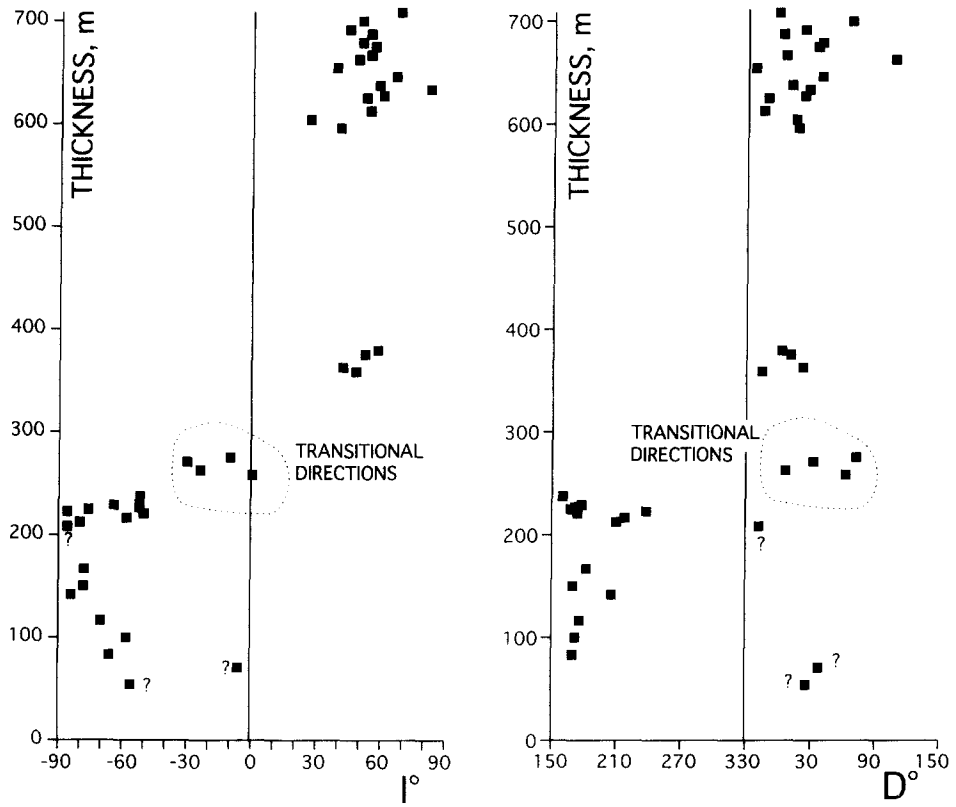


Fig. 9. Distribution of tilt-corrected ChRM declination (D) and inclination (I) through the studied stratigraphic section of the Baklan Formation (localities B1 to B6). Large gaps between the sampling points correspond to coarse-grained rocks and/or unexposed intervals.

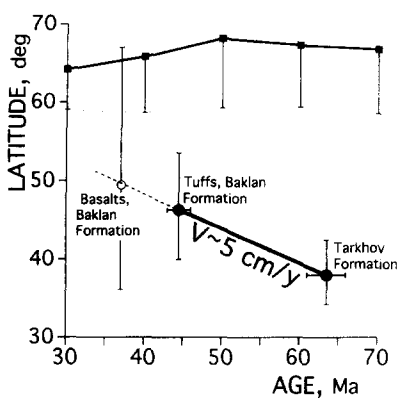


Fig. 10. Palaeolatitude versus age plot of observed values for sedimentary rocks of the Tarkhov and Baklan Formations (solid circles), basalts from the upper part of the latter formation (open circle) and the reference values for the Eurasian plate (solid squares) with associated error bars. Reference point is at 56°N 163°E. North American palaeolatitudes for the study area differ by one to three degrees from the Eurasian ones and are not shown for clarity.

the age difference between the accumulation of the Tarkhov formation and the lower part of the Baklan Formation is about 20 Ma. During the latter interval, the Kronotsky island arc moved northward by $8.9 \pm 6.3°$ with an average velocity of about 5 ± 3 cm/year. Of course, only the north component of the total movement can be deduced with the aid of palaeomagnetic data alone, and the above value should be regarded as the lower limit of the true velocity of the island arc movement.

Lithological data point to continuous volcanic activity during the accumulation of the entire Stolbov Series, including two palaeomagnetically unstudied members (the Vereschagin and Reef Formations) (Shapiro et al., 1996). Volcanic material in these two formations has both acid and mafic components, thus resembling the predominantly acid Tarkhov formation and mainly basic Baklan Formation (ibid.). Also, there are no angular unconformities within the entire Stolbov Series. Therefore, we can conclude

that the studied island arc was moving northward simultaneously with its activity and with the accumulation of the Stolbov Series.

The absolute motion of a subduction zone should have the same direction and velocity as the overriding plate (e.g., Garfunkel et al., 1986). This means that the present-day subduction zones along the northern periphery of the Pacific, e.g., the Aleutian or Kuril–Kamchatka subduction zones and related island arcs, should move southward. The established northward movement of the Palaeogene Kronotsky island arc means that this arc had not belonged to the slowly southward-moving Eurasian or North American plates with the meridional component of velocities of about 1 cm/year (Engebretson et al., 1984). At the same time, the calculated velocity of about 5 ± 3 cm/year is much less than the northern velocity component of the Kula plate (more than 10 cm/year) and matches within the error limits that of the Pacific plate (ca. 7 cm/year) (ibid.). Therefore we can assume that the Kronotsky island arc was moving with the Pacific plate since the origination in the Early Paleocene till the emplacement in the middle Miocene (Fig. 10). It also agrees with geological data which point to the Middle Miocene age of deformation and thrusting in the Tushev basin between the East Peninsulas zone and the Kamchatka mainland. This scenario fits with the available data within the uncertainties of palaeomagnetic data, kinematic parameters and palaeontological dating. This scenario in turn implies that the Kronotsky subduction zone was dipping southward. In the Early Paleocene, this subduction zone was consuming the oceanic lithosphere, most probably the oceanic periphery of a continental plate, as there are almost no traces of the Paleocene–Eocene island-arc volcanism in Kamchatka except for the studied East Peninsulas zone. This process went on from the birth of the island arc in the beginning of the Paleocene until its extinction. The time of the island arc extinction can be approximately defined from the age of the youngest rocks of the Stolbov Series island arc which is about 36–40 Ma. We can speculate that volcanism could have lasted somewhat longer then the process of subduction. In this case the subduction zone related to the Kronotsky island arc should have become extinct about 40 Ma ago. This event approximately coincides with the major reorganization of kinematics of the North Pacific 43 Ma ago.

Simultaneously with the extinction of the Kronotsky island arc, a new subduction zone responsible for Oligocene–Miocene volcanic activity in the Median Range of Kamchatka, started developing at the Pacific–Eurasia boundary. From what we know about this Oligocene–Miocene island arc, it was always on the Eurasian margin, and the Pacific plate was subducting northwestward under it. Thus the extinction of the Palaeogene island arc and formation of the other were also accompanied by change in subduction polarity.

The Kronotsky island arc was far from the Eurasia margin in the Middle Eocene. After extinction of the subduction under the Kronotsky island arc, the latter went on moving with the Pacific plate. The final collision with Eurasia occurred later, most probably in the Middle Miocene, as evidenced by deformation of the Tushev basin. A more detailed analysis of this movement will be addressed elsewhere (Levashova et al., 1997).

7. Conclusion

The main findings of our study are as follows. First, the Paleocene and Eocene palaeolatitudes corresponding to the observed inclinations of the prefolding and presumably primary magnetization components in the rocks of the island-arc complex imply that the studied island arc was originally far away from the Eurasian margin. Second, the systematic change of palaeolatitudes within the studied complex infers the steady rather rapid movement of the active island arc and related subduction zone. We assume that the oceanic crust belonging to the continental (Eurasian or North American) plate was subducting under the rapidly moving oceanic plate, most probably under the Pacific plate, and the subduction zone itself was moving together with the overriding plate. Though such a movement was predicted by theoretical studies (e.g., Garfunkel et al., 1986), to our knowledge, it is the first case in which this phenomenon is confirmed by experimental data. Of course, more data of various types are needed to better evaluate the movements of subduction zones in the Earth's history.

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